RippleLogic RLS Validation Protocol v2.6

v2.6 Current Release Integration

Release: MathGov Core Release 2026.09 v12.6 / SGP v8.5 - Independent Open-Source Qualification-Continuity Release

Exact current companion pins: Canon v12.6; SGP v8.5; ripple.md v5.5; Agent System v12.5; CSV v2.4; Cascade v2.6; Reproducibility v1.4; WDBIP v1.6; RLS Validation v2.6; Primer v4.4; Public Introduction v12.6; PC-AEP/MFDI/Source-Coupling v2.3; Aligners Sheet v5.6.

Integrity surface	Current requirement	Claim boundary
Study status	Level 1 protocol-development ready: measures, preregistration surfaces, and falsification conditions are specified.	Reliability, validity, and calibration are not established.
Latent structure	PCA, EFA, CFA, or measurement-invariance analyses require adequate samples, simulation-based planning, and preregistered interpretation criteria.	They are deferred rather than implied by a small pilot.
Workbook interface	Input validation, missingness, ranges, formulas, exclusions, and audit flags must cover the full intended data-entry region.	Workbook integrity does not validate the construct.

v2.4 Patch Note

v2.4 synchronizes normative vector headings and package references. The study design, claim boundaries, and validation hypotheses remain unchanged.

Companion validation protocol for MathGov Core Release 2026.09 v12.6 and the RippleLogic v12.6 Canon.

Status: protocol design, rater-study template, and validation sprint package. This document is not a governing core document, not empirical validation, not legal certification, and not deployment authorization.

Source binding: MathGov is the umbrella framework. RippleLogic is the decision architecture inside MathGov. The public cascade is RG -> RF -> TRC -> CSV -> RLS. The formal shorthand is RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS. RLS ranks only selectable options after Reality Grounding, the Rights Floor, the Tail-Risk Constraint, and Containment and Structural Viability have been satisfied.

Executive purpose

This protocol converts the next scientific need for RLS into a concrete study package. The goal is to test whether the 49-cell RLS welfare field is more than detailed. It must become defensible as structured, non-redundant, teachable, repeatable, and useful for better-justified decisions.

- Dimensional non-overlap question: do the 49 Scope x Dimension cells capture distinct decision-relevant information, or do some cells collapse into redundant clusters?
- Inter-rater reliability question: can trained analysts independently score the same cases with acceptable agreement?
- Decision-value question: does the RLS matrix reveal material harms, benefits, near-misses, or trade-patterns that a simpler single metric would hide?
- Revision question: which cells, anchors, instructions, or workbook fields need refinement before stronger operational-readiness claims are made?

Claim boundary

A successful validation sprint does not prove universal superiority. It can support narrower claims such as: trained raters can apply the RLS scoring manual with measurable agreement on a representative case set; some cells provide distinct information not captured by a scalar score; and identified scoring ambiguities have been logged for Canon or companion refinement. Until evidence is published or registered, the correct status is: RLS is architecturally specified and audit-ready as a structured residual welfare-ranking layer, but empirical validation of dimensional independence and rater reliability remains in progress.

The protocol does not assume that cell anchors are validated interval measurements. Reliability statistics test the repeatability and decision usefulness of bounded scoring categories under the declared method; they do not convert ordinal or bounded judgment anchors into natural units of welfare.

Canonical architecture under test

Layer	Role in validation
RG - Reality Grounding	Precondition for claim authority. Raters must not score claims stronger than the evidence surface permits.
RF - Rights Floor / NCRC	Non-compensatory gate. RLS cannot rescue rights-floor failure.
TRC - Tail-Risk Constraint	Non-compensatory catastrophic-risk gate. RLS cannot average ruin into ordinary benefit.
CSV - Containment and Structural Viability	Selectability gate for containing-system integrity and execution viability.
RLS - RippleLogic Score	Residual welfare-ranking layer over selectable options only.

Core normalized RLS formula: q(u,d)=w_u*v_d*m(u,d)*kappa(u,d), Q=sum_u sum_d q(u,d), and RLS(a)=[sum_u sum_d q(u,d)*I_prop_welfare(u,d,a)]/Q, with Q>0. If a worked example has Q=1, the numerator equals the normalized score; the denominator remains part of the canonical definition and MUST NOT be omitted from general formula surfaces.

Canonical welfare impact scale: I(u,d,a) in [-1,+1], where 0 means no material change from the declared baseline, not unknown. Unknown active cells must be marked and handled under the missing-data and gate-critical unknown rules.

v2.5 Configuration Sensitivity and Profile-First Validation

Validation cases MUST freeze the evaluated configuration before first-pass scoring. Where a materially changed configuration alters capability, evidence, controls, permissions, operating envelope, or post-state effects, it is a distinct case condition rather than an unmarked repeat.

Primary reporting remains profile-first: the 7x7 field, subgroup and worst-affected slices, uncertainty, configuration sensitivity, dominance/incomparability, and rank reversals are reported before any scalar summary. Observed successful processes are partial capability evidence and do not establish untested boundaries or transfer across configurations.

v2.3 study-integrity correction (Lineage)

The initial sprint is a burden, comprehension, boundary-classification, missingness, disagreement, and preliminary reliability study. It is not adequately powered for a defensible 49-variable factor structure merely because every cell is present. PCA, EFA, CFA, or related structure claims require a preregistered simulation-based sample justification at a later validation level.

Rater work SHALL be divided into randomized or counterbalanced blocks with recorded timing, breaks, order, and fatigue measures. Repeated quality-control cases and frozen case-packet hashes are required. First-pass welfare scoring SHALL be completed and locked before gate cues are shown; gate-cue review and adjudication occur afterward to reduce priming.

The v0.3 workbook SHALL use UNASSESSED or blank review-dependent defaults, extend validation across the entire structured range, and prevent untouched rows from appearing complete. Level 1 reporting is limited to descriptive disagreement maps, missingness, cell comprehension, preliminary reliability, boundary confusion, burden, and fatigue unless a separately preregistered design supports a stronger claim.

Study ladder

Level	Name	Minimum design	Permitted claim
L0	Internal debug	2-3 cases, 1-2 raters	Only detects obvious rubric or workbook defects.
L1	Validation sprint v0.1	10 synthetic cases, 3 options each, 3-5 raters	Exploratory evidence on clarity, disagreement, and redundancy.
L2	Calibration study v1.0	20-30 cases, 3-5 options each, 5-9 raters, mixed affiliated and non-affiliated	Preliminary IRR and dimensional-overlap evidence.
L3	External shadow-mode study	Real or historical cases, independent raters, preregistered analysis	Domain-specific evidence for operational readiness in shadow mode.
L4	Controlled pilot	Authorized institution, comparator process, outcome follow-up	Limited performance claims relative to a declared comparator.

Validation sprint v0.1

This package implements the recommended first sprint. It is intentionally small enough to run quickly and rigorous enough to reveal the main weaknesses.

- Case set: 10 realistic synthetic cases across AI governance, education, health, environment, public policy, and local infrastructure.
- Options: 3 candidate options per case.
- Raters: minimum 3, preferred 5. Include at least one non-authorial reviewer when possible.
- Scoring surface: 49 RLS cells per option, with score, confidence, rationale, anchor method, evidence basis, and uncertainty flag.
- Blind posture: raters should score independently before discussion.
- Adjudication: after independent scoring, record disagreements and produce an adjudicated reference score only after preserving raw rater scores.
- Analysis: compute IRR, disagreement heatmaps, descriptive cell correlations, missingness patterns, burden, boundary confusion, and comparison to a simpler scalar or 7-dimension summary. Do not estimate or interpret PCA, EFA, CFA, or a 49-cell latent structure at Level 1; those analyses require a later preregistered study with simulation-based sample justification.

Rater instructions

- Score the change from the declared baseline, not the absolute goodness of the option.
- Do not treat 0 as unknown. Use 0 only for no material baseline-relative change.

- If evidence is insufficient for an active cell, mark UNKNOWN_IMPACT and explain what evidence is missing.
- Do not average away subgroup harms in rights-relevant cells. Flag rights concerns separately from residual welfare scoring.
- Do not let RLS repair an option that fails RF/NCRC, TRC, or CSV.
- Record confidence independently from magnitude. A large impact with weak evidence is not the same as a small impact with strong evidence.
- Use the cell dictionary. When two cells seem similar, explain why the impact belongs in one cell, the other, or both with redundancy handling.
- When unsure, score conservatively, mark the uncertainty, and write the reviewer challenge question.

Scoring anchors

Score	Meaning	Use discipline
+1.00	Severe or maximum plausible baseline-relative benefit	Use rarely; requires strong evidence and scope-appropriate denominator.
+0.75	Major durable benefit	Large improvement with clear reach, duration, likelihood, and confidence.
+0.50	Moderate benefit	Meaningful improvement in the cell, not merely cosmetic.
+0.25	Minor benefit	Small but material benefit.
0.00	No material change from baseline	Never use as a substitute for unknown.
-0.25	Minor harm	Small but material harm.
-0.50	Moderate harm	Meaningful degradation in the cell.
-0.75	Major durable harm	Large harmful degradation, even if not gate-failing.
-1.00	Severe harm	Use rarely; often gate-relevant or escalation-worthy.

Study hypotheses

ID	Hypothesis	Evidence that supports it	Failure or revision signal
H1	RLS cells preserve non-redundant information.	Level 1: descriptive cell patterns, boundary-confusion rates, disagreement classes, and preliminary reliability show related but non-identical information. Later levels may test factor structure only under separate preregistration and design-specific sample justification.	Multiple dimensions collapse without unique decision contribution, or later preregistered structure tests fail to support the declared distinctions.
H2	Trained raters can score cells reliably.	Cell or component ICC meets target or minimum after training;	Sustained ICC below minimum after rubric training.

ID	Hypothesis	Evidence that supports it	Failure or revision signal
		disagreement is explainable and reducible.	
H3	RLS improves legibility over scalar scoring.	RLS identifies localized harms, rights-adjacent near-misses, ecological burdens, or scope-level reversals missed by scalar summary.	Scalar model produces same decision and same explanation with no material loss of information.
H4	The manual is teachable.	Raters improve after calibration and can explain cell placement consistently.	Confusion clusters persist in the same cells or dimensions.

Statistics plan

- Inter-rater reliability: use ICC(2,1) for single-rater cell scores and ICC(2,k) for averaged panel scores when the same raters score the same case-option-cell units. Use Krippendorff alpha when missingness or mixed measurement levels make ICC fragile.
- Gate and classification reliability: use percent agreement, Cohen or Fleiss kappa, or weighted kappa for binary or ordinal fields such as pass/fail, Prominent versus Guardrail, and uncertainty class. For rare failures or highly imbalanced states, percent agreement MUST NOT stand alone; report a prevalence-aware chance-corrected or class-specific measure and the underlying contingency counts.
- Ranking reliability: use Kendall W or Spearman correlation over option rankings within each case.
- Correlation analysis: build a case-option by 49-cell matrix using adjudicated or mean rater scores. Report Pearson and Spearman correlations, clustered heatmaps, and high-correlation pairs.
- Factor/PCA analysis: not part of Level 1. At later validation levels, PCA, EFA, CFA, or related latent-structure analyses MAY be run only under a separate preregistration with simulation-based sample justification, held-out checks where feasible, and explicit exploratory-versus-confirmatory labeling.
- Decision-value analysis: compare final RLS ranking and narrative justification against a simpler scalar score, a 7-dimension-only score, and a 7-scope-only score.

Provisional protocol reliability targets

Component	Target	Minimum acceptable	Protocol handling
Magnitude construction mu_k	ICC >= 0.70	ICC >= 0.60	Below minimum requires anchor/rubric redesign.
Rights-floor pass/fail	Agreement >= 0.80	Agreement >= 0.70	Below minimum requires rights mapping and subgroup guidance refinement.
TRC pass/fail	Agreement >= 0.80	Agreement >= 0.70	Below minimum requires scenario and probability guidance refinement.
RLS ranking order	Agreement >= 0.70	Agreement >= 0.60	Below minimum requires score,

Component	Target	Minimum acceptable	Protocol handling
			uncertainty, or tie-break review.
PLSS prominence classification	Agreement >= 0.70	Agreement >= 0.60	Below minimum requires prominence-signal guidance refinement.

Correlation and factor-analysis revision rules

- If two cells correlate above 0.85 across most domains and raters cannot explain distinct causal meanings, mark the pair for overlap review.
- If a cell has persistently low variance across diverse cases, check whether it is genuinely rare, poorly defined, over-masked, or under-evidenced.
- If a dimension shows repeated cross-loading with another dimension, add boundary examples to the scoring manual before considering structural revision.
- If a cell has low reliability but high decision relevance, do not remove it prematurely. Strengthen anchors, examples, evidence requirements, and reviewer checks first.
- If simplified models produce the same winner but lose key localized risk signals, preserve RLS and report the extra legibility as decision-relevant detail.
- If simplified models produce the same winner and same explanation with no material information loss across a larger case set, consider a reduced mode or domain-specific profile.

Disagreement taxonomy

Code	Meaning	Likely fix
DGT_SCOPE	Raters disagree on Union Scope mapping.	Improve stakeholder-instance and scope mapping examples.
DGT_DIMENSION	Raters disagree on welfare dimension.	Add boundary examples between dimensions.
DGT_MAGNITUDE	Raters agree direction but not size.	Improve score anchors and indicator scaling.
DGT_SIGN	Raters disagree benefit versus harm.	Clarify baseline, causal pathway, or evidence surface.
DGT_CONFIDENCE	Raters agree magnitude but not confidence.	Improve evidence-quality mapping.
DGT_GATE	Raters disagree on rights, TRC, or CSV relevance.	Strengthen gate trigger guidance.
DGT_UNKNOWN	Raters disagree whether evidence is sufficient.	Clarify unknown and phantom-instance handling.
DGT_DOUBLECOUNT	Raters place same effect in multiple cells inconsistently.	Add redundancy-handling examples.

Case packet standard

- Decision boundary and baseline.
- Option set with 2-5 options.
- Stakeholder-instance map and active Union Scopes.
- Evidence packet, including what is known, unknown, contested, and assumed.
- Rights Floor, TRC, and CSV pre-screen notes.

- RLS scoring sheet for every option.
- Rater confidence and rationale fields.
- Adjudication sheet that preserves raw scores before consensus.
- Post-analysis revision notes.

Synthetic case roster

Case	Title	Purpose
C01	AI tutor in public schools	Compare mandatory deployment, opt-in teacher-supervised deployment, and no deployment for a public-school AI tutor.
C02	Flood relocation policy	Compare voluntary relocation support, mandatory relocation, and infrastructure-only flood protection for a high-risk community.
C03	Agricultural pesticide decision	Compare current pesticide use, restricted integrated pest management, and rapid ban with transition subsidy.
C04	Hospital triage support system	Compare clinician-only triage, advisory AI triage, and automated triage for non-emergency scheduling.
C05	Social-media moderation rule	Compare minimal moderation, rights-protecting graduated moderation, and aggressive automated removal.
C06	Welfare fraud detection model	Compare manual review, AI risk flagging with human appeal, and automated benefit suspension.
C07	Urban traffic redesign	Compare car-priority road widening, bus/bike corridor redesign, and congestion pricing with equity rebates.
C08	Renewable microgrid project	Compare no project, community-owned solar microgrid, and vendor-owned microgrid with long lock-in contract.
C09	School phone policy	Compare unrestricted use, blanket ban, and structured phone zones with exceptions and student voice.
C10	Local data center proposal	Compare approval as proposed, approval with water/energy/community safeguards, and rejection pending regional capacity review.

49-cell welfare dictionary - compact validation version

This compact dictionary is extracted against the Canon Appendix AD role: it supports scoring interpretation and reviewer literacy. It does not modify gates, thresholds, equations, or claim boundaries.

Cell	Label	Plain meaning	Reviewer check
U1/D1	Personal Resources	An individual's economic security: income, savings, housing stability, and access to the material means of a decent life.	A reviewer checks whether the income/housing change is measured against a declared baseline cohort, not anecdote.
U1/D2	Personal Health	The individual's physical and mental health, safety, and bodily integrity.	A reviewer asks whether a claimed health gain rests on outcome data or only on inputs (e.g. clinics built ≠ health improved).
U1/D3	Relationships	The quality and stability of a person's close personal relationships and social connection.	A reviewer challenges whether a relationship claim is evidenced or inferred from a proxy like attendance.
U1/D4	Learning	The individual's access to education, skills, and the capacity to learn and develop.	A reviewer asks whether learning outcomes are demonstrated or only opportunity offered.
U1/D5	Personal Agency	The person's real ability to choose, refuse, consent, act, and shape their own life path.	A reviewer tests whether consent was genuine and revocable, or merely formal.
U1/D6	Purpose	The individual's sense of meaning, dignity, and worth in their life and work.	A reviewer distinguishes a genuine dignity effect from sentiment unsupported by experience data.
U1/D7	Local Conditions	The immediate physical environment a person lives in: air, noise, water, hazards, and surroundings.	A reviewer checks measured exposure change against the affected location, not a city-wide average.
U2/D1	Household Resources	The economic security of the household unit: shared income, assets, housing, and ability to meet needs.	A reviewer asks whether the household, not just the earner, is the measured unit.
U2/D2	Household Health	The collective physical and mental health and safety of	A reviewer checks that dependents and caregivers are scored

Cell	Label	Plain meaning	Reviewer check
		household members, including dependents.	separately, not blended into one household figure.
U2/D3	Family Cohesion	The strength, stability, and supportive quality of relationships within the household.	A reviewer asks whether a cohesion claim rests on outcomes or on assumed effects of a program.
U2/D4	Shared Learning	The household's collective access to information, education, and learning capacity.	A reviewer checks access actually reaches the household, not just the area.
U2/D5	Household Agency	The household's collective ability to make decisions, plan, and control its own circumstances.	A reviewer tests whether household choice is real or constrained by conditions.
U2/D6	Family Meaning	The household's shared sense of identity, belonging, dignity, and purpose.	A reviewer separates a real meaning effect from program rhetoric.
U2/D7	Home Environment	The physical environmental quality of the home and its immediate setting.	A reviewer checks measured home conditions, not self-report alone.
U3/D1	Local Resources	The shared economic resources, infrastructure, and services available to a local community.	A reviewer asks whether the benefit is genuinely shared or captured by a subgroup.
U3/D2	Community Health	The collective physical and mental health and safety of the local population.	A reviewer checks whether subgroup harms are masked by a favourable community average.
U3/D3	Community Cohesion	The quality of trust, cooperation, belonging, safety, and relational fabric within a community.	A reviewer asks whether cohesion gains for some came via exclusion of others.
U3/D4	Local Knowledge	The community's shared knowledge, information access, and local expertise.	A reviewer checks consent and ownership of the knowledge claimed as preserved.
U3/D5	Community Agency	The community's collective capacity to organise, participate, and influence decisions affecting it.	A reviewer tests whether participation was substantive or a procedural formality.
U3/D6	Shared Meaning	The community's	A reviewer separates

Cell	Label	Plain meaning	Reviewer check
		shared identity, culture, dignity, and sense of collective purpose.	a genuine cultural effect from a symbolic gesture.
U3/D7	Local Ecology	The ecological condition of the community's local environment: green space, biodiversity, water, land.	A reviewer asks whether 'no local effect' was measured or merely assumed.
U4/D1	Organizational Resources	An organization's operating means: finances, capital, staffing, and capacity to function and deliver.	A reviewer checks whether the organizational gain creates uncosted externalities in other cells (the classic 'operating resources up, U7 down' trap).
U4/D2	Organizational Safety	The safety of people within and affected by the organization: workers, users, and the public.	A reviewer checks whether reported safety reflects outcomes or only paperwork compliance.
U4/D3	Organizational Culture	The internal relational health of the organization: trust, fairness, inclusion, and cooperation.	A reviewer asks whether culture data is independent or self-reported by leadership.
U4/D4	Knowledge Systems	The organization's knowledge, data integrity, institutional memory, and learning systems.	A reviewer tests whether claimed integrity is verifiable or asserted.
U4/D5	Role Agency	The meaningful autonomy, voice, and fair treatment of individuals in their organizational roles.	A reviewer checks whether voice mechanisms produce outcomes or are decorative.
U4/D6	Mission Coherence	The alignment between the organization's stated purpose and its actual conduct and effects.	A reviewer asks whether coherence is evidenced by conduct or only by restated values.
U4/D7	Operational Footprint	The environmental impact of the organization's operations: emissions, waste, resource use, land.	A reviewer checks footprint against measured externalities, not offset claims alone.
U5/D1	Public Resources	The polity's fiscal and material capacity: public finances, infrastructure, and	A reviewer checks whether gains are sustainable or borrowed from the

Cell	Label	Plain meaning	Reviewer check
		provisioning.	future.
U5/D2	Population Health	The health, safety, and mortality outcomes of the whole population within a polity.	A reviewer checks whether worst-case scenarios were bounded with a tail measure, not averaged away.
U5/D3	Legitimacy	The rule of law, due process, accountability, and perceived legitimacy of public institutions.	A reviewer tests whether legitimacy is measured independently or claimed by the institution being assessed.
U5/D4	Information Access	The polity's information environment: access, transparency, free expression, and epistemic integrity.	A reviewer checks whether transparency is real and usable or nominal.
U5/D5	Civil Agency	People's ability to participate in public life, exercise rights, contest authority, and influence governance.	A reviewer tests whether civic participation is enforceable or merely nominal, and whether any group is selectively excluded.
U5/D6	Public Meaning	Shared civic identity, dignity, social contract, and non-domination across the polity.	A reviewer separates symbolic recognition from changes people actually experience.
U5/D7	Public Environment	The environmental quality and ecological sustainability managed at the polity scale.	A reviewer checks whether environmental-justice distribution was assessed, not just regional totals.
U6/D1	Global Resources	Humanity's shared material base: global resource stocks, critical supply chains, and common infrastructure.	A reviewer checks whether 'global' claims rest on global data or extrapolated local figures.
U6/D2	Human Safety	The safety and survival of humanity at large, including existential and catastrophic risk.	A reviewer checks that tail risk was bounded with a worst-case measure and not diluted by expected-value framing.
U6/D3	Cooperation	The capacity of humanity to coordinate, cooperate, and	A reviewer asks whether cooperation is durable or a fragile short-term

Cell	Label	Plain meaning	Reviewer check
		maintain peaceful, stable relations at scale.	arrangement.
U6/D4	Civilizational Knowledge	Humanity's accumulated knowledge, science, and the integrity and safety of its knowledge systems.	A reviewer checks whether knowledge benefits were weighed against misuse and proliferation pathways.
U6/D5	Collective Agency	Humanity's capacity for self-determination and legitimate collective decision-making at the species scale.	A reviewer tests whether 'collective' agency includes the marginalised or only powerful actors.
U6/D6	Shared Purpose	Humanity's shared sense of meaning, moral direction, and long-term orientation toward flourishing.	A reviewer flags this cell as especially prone to rhetoric and demands concrete mechanisms.
U6/D7	Planetary Conditions	The condition of planetary systems that support human civilisation and coordinated managing intelligence.	A reviewer checks whether irreversibility and worst-case bounds were modelled, not just expected outcomes.
U7/D1	Life-Support Resources	The biosphere's foundational resources that sustain life: soil, fresh water, clean air, fertility.	A reviewer checks whether renewal rates, not just current stocks, were assessed.
U7/D2	Ecosystem Health	The health, function, and viability of ecosystems and the species within them.	A reviewer checks whether ecosystem-function evidence exists or only a single charismatic indicator.
U7/D3	Biotic Relations	The integrity of relationships and interdependencies among living systems and between humans and nature.	A reviewer asks whether relational/cascade effects were modelled or ignored as 'no direct effect'.
U7/D4	Ecological Knowledge	Humanity's understanding and monitoring of ecological systems, including baseline and early-warning data.	A reviewer checks that absent ecological data is marked unknown, not scored as 0 ('no effect').
U7/D5	Resilience Capacity	The biosphere's capacity to absorb shocks, adapt, and	A reviewer asks whether proximity to thresholds was

Cell	Label	Plain meaning	Reviewer check
		recover from disturbance.	assessed, not just current condition.
U7/D6	Life Continuity	The continuity and persistence of life and biodiversity over time, including irreversibility of loss.	A reviewer checks whether irreversibility was treated as a tail constraint, not a discountable cost.
U7/D7	Ecological Integrity	The overall health, resilience, diversity, and continuity of ecosystems as living support systems.	A reviewer checks whether integrity is evidenced by ecosystem-level data or inferred from a single metric.

Recommended additions to the release ecosystem

Add the following as companion, non-governing materials. They should be versioned separately from the core canon unless a future release explicitly promotes part of them into normative core.

Artifact	Role
RippleLogic_RLS_Validation_Protocol_v2_6	Study protocol, rater instructions, hypotheses, analysis plan, and revision rules.
RLS_Validation_Workbook_v0.3	Rater-entry and analysis-prep workbook.
RLS_Calibration_Note_v1.0	Future results document after the first scoring sprint.
Case_Packets/	Folder for frozen synthetic and real shadow-mode case packets.

Proposed README wording

Current status: RLS is specification-ready and audit-ready as a structured residual welfare-ranking specification and audit method. Empirical validation of dimensional independence, inter-rater reliability, and decision-performance advantage is a next-stage research priority. The RLS Validation Protocol provides the study design for testing these claims through scored cases, independent raters, reliability statistics, and correlation/factor analysis of the 49-cell welfare field.

Completion definition for the first sprint

- At least 10 cases scored independently by at least 3 raters.
- All raw rater scores preserved.
- Completion, missingness, and disagreement statistics produced.
- IRR reported for score, ranking, and key classifications where applicable.
- Correlation or redundancy heatmap produced over the 49 cells.
- At least one scalar or reduced-matrix comparator run.
- Revision log produced with exact cells, anchors, or instructions needing repair.
- Public claim boundary updated: exploratory, preliminary, validated for limited domain, or not validated.

Physical-admissibility validation extension

The v12-line research ladder, introduced at v12.0 and retained in the current v12.6 release, treats physical admissibility claims as a separate validation surface from governance mandate and authorization. Future validation runs should measure whether reviewers can reliably distinguish: governance authorization only, physical admissibility supported within the declared validity domain, physical admissibility not established, and physical admissibility contraindicated.

This extension does not test whether MathGov computes physics. It tests whether MathGov records the correct evidence boundary and refuses to overclaim physical safety when the domain evidence is missing or insufficient.

Minimum successful first validation report

A first RLS validation report should publish, at minimum:

Required output	Purpose
Raw rater score CSV	Allows independent review of cell-level ratings and missingness.
Adjudication log	Shows how disagreements, uncertainty, and gate-relevant disputes were handled.
ICC / alpha results	Reports inter-rater reliability and internal consistency where appropriate.
Disagreement heatmaps	Reveals unstable cells, dimensions, or union scopes.
Revision register	Records proposed changes to definitions, examples, thresholds, or training material.
Claim-boundary update	States what the results do and do not validate.

v12.6 mandatory conformance vectors

Single-source and vector mapping rule (Normative for protocol conformance).

RLSV extension family	Canon control / mapping	Purpose
RLSV-IRR-*	Canon §17.4A	Reliability and disagreement studies.
RLSV-RANK-*	Appendix R decisiveness, masking, and non-selectability vectors	Ranking and refusal behavior.
RLSV-DEP-*	Appendix R.20 dependence-sensitivity stress	Dependence-cluster robustness.
RLSV-BOUNDARY-*	Appendix R category, rights, tail, CSV, and claim-boundary vectors	Construct and interface discrimination.

Canon §17.4A controls IRR targets and Canon Appendix R controls framework conformance vectors. This protocol operationalizes those targets. Protocol-specific extensions use the RLSV- prefix and MUST state the Appendix R vector or invariant they extend; they do not create an independent framework-conformance authority.

The validation suite must include at least the following deterministic conformance cases before empirical claims are considered:

1. one-month categorical bodily-integrity violation: RF/NCRC must fail regardless of short duration;
2. one-year arbitrary detention: RF/NCRC must fail or return unknown/escalate if the categorical profile is unresolved;
3. low-probability lethal exposure: a severe-hazard probability bound and governed tolerance are required; missing fields cannot pass;
4. every option fails TRC while safe delay is available: Tail Emergency Mode must not activate;

5. every option fails TRC in a demonstrated emergency: only a below-absolute-cap provisional least-CVaR option may proceed, with no ordinary RLS;
6. non-default κ values that would produce an unnormalized raw score above 1: normalized RLS must remain within [-1,+1];
7. active masks that change effective weight mass: score scale must remain invariant under normalization;
8. uncertain equal-and-opposite impact instances: Method B uncertainty must remain positive through pre-cancellation contribution mass.

Current-package Tier-Integrity Conformance Vectors (Normative for protocol conformance)

The validation package MUST include the following cases:

1. TIER2_CSV_REQUIRED_BEFORE_RLS: an option passes NCRC and TRC but has no CSV status. Expected: no ordinary RLS selection claim; route to CSV review or refusal.
2. TIER2_CSV_NOT_MATERIAL_WITH_RATIONALE: a low-structural-materiality option records the rationale and may enter RLS.
3. TIER2_CSV_REDESIGN_REQUIRED_BLOCKS_RLS: an option with CSV_REDESIGN_REQUIRED is excluded before scoring.
4. TRC_ALL_FAIL_SAFE_DELAY: expected redesign/delay/refusal; minimum CVaR does not create selection.
5. TAIL_EMERGENCY_PREREQUISITE_MISSING: expected TAIL_EMERGENCY_REFUSED or escalation.
6. TAIL_EMERGENCY_FULL_PREREQUISITES: minimum-CVaR provisional action is allowed only inside Tail Emergency Mode, with ordinary RLS disabled.
7. RIGHTS_UNKNOWN_NOT_ZERO: missing material rights evidence returns an unknown/refusal posture rather than a zero impact.
8. RLS_ZERO_ACTIVE_MASS: returns RLS_NO_ACTIVE_MASS, not zero.

Appendix: Current-package source-integrity vectors

ID	Input	Required result
MPS-HYPOTHESIS-01	RLS ranking changes between h=0 and h=1 for a non-FPP stakeholder with decision-material MPS evidence.	MPS_HYPOTHESIS_SENSITIVE; no convenient intermediate coefficient; preserve multiple options, seek evidence, apply precaution, narrow, or refuse.
MPS-NE-01	Target boundary is valid but admissible evidence is insufficient or observability is poor.	MPS-NE; no conversion to MPS-0, zero, or a scalar multiplier.
UCI-NA-01	U1 Equity is inapplicable; H=F=R=0.	E is NA, remaining weights renormalize, UCI=0; no fixed perfect equity value.
UCI-UNKNOWN-01	A governed within-self equity instrument is applicable but evidence is missing.	E is UNKNOWN and the claim is narrowed/escalated; it is not NA or 1.
RIGHTS-HAZARD-01	Credible fatality pathway with intrinsic severity 0.80 and non-negligible exposure.	Severe-rights-hazard channel activates independently of LIFE floor threshold 0.90.
GATE-BOUND-01	Low-confidence adverse catastrophe or material CSV	Apply the deterministic adverse bound; if not

ID	Input	Required result
	instance could change a gate.	reproducible, return UNKNOWN/ESCALATE/NARROW/REFUSE, never a confidence-discounted pass.
RLS-FULL-MASK-01	Eight nonzero cells, all 49 cells active with explicit evidence-backed zeros.	Q=1 and normalized RLS equals the contribution sum.